

Assignment 5 - Unconsidered Metrics

The Vital sign I think can be added in the data set is Peripheral oxygen saturation. SpO₂ is a vital sign that measures the percentage of hemoglobin in the blood carrying oxygen, providing a quick, non-invasive snapshot of how effectively the heart and lungs are delivering oxygen to the body. Normal SpO₂ values range between 95–100%, which indicates adequate oxygen level. Readings of 90–94% suggest mild hypoxia, 85–89% indicate moderate hypoxia, and values below 85% reflect severe hypoxia requiring immediate intervention. In the triage process, SpO₂ is significant because it serves as an early warning marker for respiratory or circulatory compromise. By identifying abnormal oxygen levels before more obvious symptoms appear, clinicians can prioritize care, escalate treatment, and prevent life-threatening deterioration. This makes SpO₂ not only a routine measurement but also a critical tool for rapid decision-making in emergency and clinical settings.

SpO₂ readings can sometimes be misleading because certain factors interfere with the accuracy of the measurement. Cold hands, poor circulation, or patient movement can cause the device to misread oxygen levels. Nail polish, artificial nails, or even strong ambient light may block or distort the sensor's signal. In addition, conditions like carbon monoxide poisoning can falsely elevate SpO₂ values since the device cannot distinguish between oxygen and other gases bound to hemoglobin.